Jonguk Cheong / Ph.D. Student

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About Me

I am a Ph.D. student in the CompSec Lab at Seoul National University, advised by Prof. Byoungyoung Lee. My research interests lie in systems security, with a focus on LLM serving systems and confidential computing. Most recently I built **KVGuard**, a sanitizer that finds memory-safety bugs in the KV cache of LLM serving engines such as vLLM and SGLang.

Education

- **Seoul National University** Mar. 2026 – Present
  Seoul, South Korea
  Ph.D. in Electrical and Computer Engineering (Advisor: Byoungyoung Lee)
- **Seoul National University** Mar. 2020 – Feb. 2026
  Seoul, South Korea military service (2 years)
  B.S. in Electrical and Computer Engineering

Publications

- **KVGuard: A KV Cache Safety Checker for LLM Serving Systems**
  Jonguk Cheong, Myungbin Hwang, Eulgyu Kim, and Byoungyoung Lee
  arXiv preprint, 2026.

Talks

- **KVGuard: A KV Cache Safety Checker for LLM Serving Systems** Sep. 2026
  Intel Academic Security Conference, Hillsboro, OR, USA

Projects

- **KV cache sanitizer for LLM serving systems** Apr. 2026 – Present
  - Designed a sanitizer that detects memory-safety bugs in the KV cache of real-world LLM serving systems
  - **Found 8 new KV cache memory-safety bugs in vLLM and SGLang**
- **Emulating Intel TDX machines** Aug. 2025 – Present
  - Built a full-system emulation framework for Intel TDX with KVM
- **Protecting Linux kernel code mappings with Intel VT-rp** Aug. 2024 – Dec. 2025
  - Added Intel VT-rp support to KVM and the guest Linux kernel, which upstream Linux does not provide
  - Used it to protect kernel code mappings against an attacker with full guest kernel privilege

Technical Skills

- **Languages:** C, C++, Python, CUDA
- **Systems:** Linux kernel, KVM, QEMU, Intel TDX, vLLM, SGLang
- **Tools:** AFL, Syzkaller, Git, Docker